ANKUR KUMAR SINGH

+91-8891079725 • ankurrajput7050@gmail.com
GitHub: github.com/Ankur377823 • LinkedIn: linkedin.com/in/ankur-singh0112 • Portfolio: portfolio-puce-nine-11.vercel.app

PROFESSIONAL SUMMARY

Self-taught software engineer pursuing a B.Tech at NIT Calicut, currently interning at Wingback Security on AI security and LLM guardrail systems. Skilled in Python, SQL, FastAPI, and ML tools like TensorFlow, PyTorch, and Scikit-Learn. Solved 800+ DSA problems on LeetCode. Looking for software engineering, backend, or applied AI/ML roles.

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL
ML & AI: Scikit-Learn, TensorFlow, Keras, PyTorch, XGBoost, YOLO, OpenCV, SHAP, NumPy, Pandas
Backend & APIs: FastAPI, REST API, Pydantic, Uvicorn, LiteLLM
Data & Databases: SQL, MongoDB, query design, data pipelines
DevOps & Tools: Docker, Kubernetes, Git, GitHub, Linux, Jupyter Notebook
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Problem Solving

EXPERIENCE

Software Engineering Intern — Wingback Security **June 2026 – Present**

- Built a Python FastAPI tool that connects LiteLLM to Wingback's AI threat detection system, checking AI inputs and outputs for issues like PII, secrets, and prompt injection across OpenAI, Anthropic, and other providers, with no changes needed to customer code.
- Designed checks that run before and after each AI request, with monitor and enforce modes, to keep AI systems safe and reliable in production.
- Built structured logs (without storing raw data) to support monitoring and responsible AI practices, and used SQL to check and validate log data while debugging.
- Wrote 32 automated tests using pytest and respx to check different response types (blocked, allowed, flagged), timeout handling, and full request flows before deployment.

AI/ML Team Member — Google Developer Student Club (GDSC), NIT Calicut **Jan 2025 – Present**

- Contributed to ML project deployments and researched new AI/ML trends such as LLMs, computer vision, and model serving.
- Worked with student teams to build and improve machine learning solutions for real-world problems.

PROJECTS

Botpress Chat API Connector — Red-Team Scanning Demo
Python, FastAPI, SQLite, Botpress Cloud

- Built a connector for Botpress Cloud bots that sets up a bot, checks its connection, and tests it with tricky prompts (like prompt injection, PII leaks, and jailbreaks), recording each response, response time, and other details.
- Built a connector library that works with the live Botpress Chat API and stores scan history in SQLite for each bot so results can be reviewed later.
- Wrote 26 automated tests using pytest and a mock Botpress server to check connection validation, scan runs, timeout handling, and the full onboard-to-history process.

Car Price Prediction API
Python, FastAPI, Scikit-Learn, Redis, Docker, Prometheus

- Trained a Random Forest model with Scikit-Learn and built a FastAPI REST API to serve price predictions, with JWT authentication and API key checks.
- Added Redis caching to avoid repeating the same predictions for repeated requests.
- Packaged the full stack (FastAPI, Redis, Prometheus, Grafana) with Docker Compose and deployed it on Render with managed Redis to monitor performance in real time.

EDUCATION

National Institute of Technology Calicut **2023 – 2027**
B.Tech in Civil Engineering | CGPA: 8.24 / 10.0

Daya Prakash Saraswati Vidya Mandir **2022 – 2023**
Intermediate | 87.4%

Keshav Saraswati Vidya Mandir **2020 – 2021**
Matriculate | 82.2%

ACHIEVEMENTS

- National Semifinalist – Flipkart GRiD 8.0: reached the national semifinal round of Flipkart's engineering challenge.
- 2nd Place – Yuva Manthan Hackathon (2024): recognized for problem solving, teamwork, and coding under pressure.
- 800+ DSA problems solved on LeetCode, across arrays, graphs, trees, and dynamic programming.